

Shiyu Chen 陈时宇

Pittsburgh, PA · +1 626-740-4571 · shiyuche@andrew.cmu.edu · chencn22.github.io

Research interests: autonomous exploration & active perception · semantic mapping & LiDAR–camera fusion · next-best-view planning · VLM-in-the-loop industrial inspection

EDUCATION

Carnegie Mellon University — M.S. in Mechanical Engineering (Research) Aug 2025 – May 2027 (expected)
Computational Engineering and Robotics Lab (CERLAB), advised by Prof. Kenji Shimada Pittsburgh, PA

- Focus: 5-DoF semantic exploration and active inspection for legged robots in industrial environments.

Univ. of Cincinnati & Chongqing University — Joint Program, B.S. / B.Eng. in Mechanical Engineering 2020 – 2025

Robotics Minor (UC) · UC GPA 3.71/4.0 · CQU GPA 3.58/4.0 · Dean's List ×4 Cincinnati, OH / Chongqing, CN

RESEARCH EXPERIENCE

Graduate Researcher, CERLAB Roboteam (CMU × YKK factory-automation inspection) Aug 2025 – Present

5-DoF semantic exploration with VLM-guided active inspection (co-first author · arXiv:2609.19460 · submitted to ICRA 2027)

- Spot tilts its body (pitch, roll) to see the upper surfaces a level camera misses — **5-DoF viewpoints** on top of planar motion.
- Semantic mapping: per-instance detection masks back-projected into 3D object boxes in the occupancy map.
- Viewpoint sampler picks body postures by expected coverage gain; aim-aligned execution avoids needless reorientation.
- **VLM-assisted viewpoint pruning** from observation history + bird's-eye-view map; geometric fallback so the stack never blocks.
- Built the Isaac Sim benchmark (three industrial scenes, ground-truth surfaces, plug-in planners, Dockerized; C++/ROS planner).
- **+8–10 pp** coverage and **17–32% less exploration time** vs. the planar 3-DoF baseline; **53–73% fewer postures** than competing 5-DoF methods; highest mean object-coverage AUC.
- Demonstrated on a real Spot in a machine shop (five target machines).

Real-time LiDAR–camera semantic registration on a legged platform

- Fused Ouster LiDAR geometry with Insta360 X4 panoramic imagery on a Boston Dynamics Spot.
- **10 FPS** at full LiDAR rate, **<100 ms** latency, **<10 px** reprojection error.
- Kept registration consistent under gait vibration; designed a 3D-printed rigid sensor mount for stable extrinsics.

Undergraduate Researcher, Cable-Driven Soft Robotic Arm for Minimally Invasive Surgery Sep 2024 – May 2025

Univ. of Cincinnati, advised by Prof. Janet (Jiangxia) Dong

- Designed and prototyped a wire-actuated continuum arm: **0.27 mm/step** resolution, **>230°** bending range.
- Modeled nonlinear actuation; led hardware–control integration with clinical end-users; published in *ASME J. Medical Diagnostics* (2026).

Research Assistant, Deep-Learning Defect Detection in Composite Materials Jan 2022 – Jul 2022

Chongqing University, advised by Prof. Bo Yang

- CNN models (LeNet/VGG16) with multi-scale feature extraction for defect detection in carbon-fiber and welded parts.
- Reached **>97%** accuracy; applied CenterNet/EfficientDet for defect localization.

PUBLICATIONS & PATENTS

X. Zhan^{*}, **S. Chen^{*}**, K. Shimada, "Pose-aware Legged Robot Semantic Exploration with Omnidirectional Perception in Confined Unknown Environments," *arXiv:2609.19460*, Sep 2026. Submitted to ICRA 2027 (under review). ^{*}Equal contribution.

Y. Wang, C. Liu, K. Jiang, B. Wu, **S. Chen**, J. Dong, A. Ashfaq, "Innovative Flexible Robotic Arm for Enhanced Precision in Transaortic Surgical Myectomy," *ASME Journal of Medical Diagnostics*, 9(2): 021001, 2026.

L. Liu, **S. Chen**, Z. Tang, "Intelligent Diagnosis of Bearing Faults Based on SDP Image Fusion," *Int. Conf. on Image Processing, Computer Vision and Machine Learning (ICICML)*, Chengdu, 2023.

Patents (China, as co-inventor): 4 granted invention patents (2025–2026) on deep-learning machine vision for steelmaking process monitoring (steel-stream landing-point, slag-splashing and slag-discharge detection); 3 pending applications in belt-conveyor inspection.

INDUSTRY EXPERIENCE

Computer Vision R&D Assistant, CISDI Information Technology May 2022 – Dec 2023
Chongqing, China

- Trained and optimized PyTorch detection models for an industrial belt-conveyor inspection system.
- Deployed to edge devices (NVIDIA Xavier NX / AGX Orin) via TensorRT, cutting inference latency by **40%**; ported Python to C++.
- Adapted models to Huawei Ascend (2nd prize, Ascend AI competition); built a PyQt annotation tool for 100k+ images. Return offer.

TECHNICAL SKILLS

Programming	C++ (proficient), Python (proficient), MATLAB, C
Robotics / Sim	ROS1 & ROS2, NVIDIA Isaac Sim, Spot SDK, LiDAR–camera calibration, frontier exploration, next-best-view / active perception, SLAM, path planning (TSP/LKH)
Perception / ML	PyTorch, TensorRT, YOLOv7, SAM2, VLM APIs, CNN/deep learning, point-cloud processing (ICP)
Tools	Linux, Docker, Git, SolidWorks, COMSOL

HONORS & TEACHING

- **National Bronze Prize**, 8th China Int'l "Internet+" Innovation & Entrepreneurship Competition, 2023.
- Teaching Assistant — Engineering Design Thinking II (UC, 2025) and C++ Programming (CQU, 2024).